Online PES-Adapted Neuromorphic Membrane–Gate Residual Control for Actuator-Constrained UAV Quaternion Attitude Tracking

***Abstract*—We study a compact neuromorphic residual architecture for quaternion attitude tracking with bounded actuator commands. A fixed quaternion proportional–derivative (PD) controller is augmented by six paired, continuous coupled membrane–gate (CMG) units. Their gate-derived activities are decoded through eighteen weights adapted online by prescribed error sensitivity (PES). We distinguish intrinsic conductance modulation, decoder learning, and residual authority, and establish bounded neural states, bounded command deviation, projected decoder boundedness, and a restricted frozen-feature learning result. These properties do not imply closed-loop stability or saturation-free rotor allocation. In ten matched 45-s MuJoCo cases spanning five missions, an X2 model and an explicitly labeled enhanced-propulsion CF2 design exhibit 17.3–60.6% lower mean quaternion-angle error than PD. Actuator-bound occupancy is zero except for brief Hardstep intervals. Direct and Nengo realizations agree within 2.013×10^{-6} in relative mean-error difference. Ablations identify online decoder adaptation, rather than conductance modulation, as the principal source of improvement in the tested amplified-gate regime. An annealed activity-trace learning variant is examined separately in the precursor architecture, with both beneficial and adverse outcomes. Preliminary physical 3DOF Hover Roll tests provide supporting evidence: three matched pairs improve their recorded mean error by $2.65\% \pm 1.08\%$ (sample standard deviation).**

I. Introduction

Learning-based augmentation can retain useful classical feedback structure while adapting a correction that is difficult to specify a priori. Residual robot control is established [1]; quadrotor examples include geometric $\mathcal{L}_1$ augmentation with performance guarantees [2] and PID-plus-PPO attitude residuals [3]. Thus, neither additive residual learning nor online flight adaptation is novel in isolation.

Neuromorphic control offers a complementary implementation approach. Nengo supports the Neural Engineering Framework (NEF) and online decoder learning [4]. PES-based adaptive control has been demonstrated on manipulators [5], [6], and a Nengo spiking PID controller with PES-based self-tuning has been validated on a free-flying UAV [7]. Our contribution is not the first PES/Nengo UAV controller. Instead, we investigate a specific continuous membrane–gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

The central question is whether this small stateful basis can improve tracking, and which of its components explains the improvement. The paper contributes: (i) a code-aligned CMG–PES architecture with paired currents, explicit state/readout equations, and actuator interfaces; (ii) precise boundedness and baseline-recovery properties, without an unsupported closed-loop stability claim; (iii) a five-mission, two-model study with direct/Nengo agreement and neuron-family comparisons; and (iv) a separate study of annealed activity-trace learning that documents its implementation and evidence limitations. Selected preliminary Hover tests support physical feasibility but are not the main validation claim.

Here "neuromorphic" refers to neuron-inspired continuous state dynamics, local error-directed learning, and a neural-simulator realization. The implementation does not emit spikes and has not been deployed on neuromorphic hardware; no power advantage is claimed. PES is an established rule, and novelty is claimed for its integration and controlled evaluation in this architecture, not for PES itself.

II. Attitude Model, Baseline, and Actuator Interface

A. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ be the scalar-first Hamilton quaternion mapping body coordinates to inertial coordinates, and let $\boldsymbol{\omega}$ be the body angular velocity. The rigid-body equations are

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}), \qquad \boldsymbol{J} \succ 0. \tag{2}$$

The archived MuJoCo implementation conjugates its reported orientation: the controller uses the passive quaternion $q = Q^{-1}$, hence

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^{\mathsf{T}}]^{\mathsf{T}} \otimes q. \tag{3}$$

The desired q_d uses the same convention. With $R(q)$ denoting the rotation represented by a quaternion, define

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^{\mathsf{T}}]^{\mathsf{T}}, \tag{4}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0})\boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2\,\mathrm{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The last expression is the shortest rotation angle; it is unchanged by replacing q_e with $-q_e$. The sign-selected error has a discontinuity at $q_{e,0} = 0$. No robustness at that switching

surface or hybrid global stability is inferred from static sign selection [8].

If q_d obeys (3) with desired-frame rate $\boldsymbol{\omega}_d$, direct differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^{\mathsf{T}}]^{\mathsf{T}}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^{\mathsf{T}} \boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2} \boldsymbol{q}_{e,v}^{\mathsf{T}} \boldsymbol{\omega}_e. \tag{8}$$

This identity fixes the frame and sign convention; it is not a convergence claim. Reference Euler triples below parameterize q_d in this passive convention. They must not be silently interpreted as active body-to-world Euler commands.

B. Fixed Quaternion PD Baseline

With componentwise multiplication $\odot$, the implemented law is

$$\boldsymbol{k}_q = 2\,\mathrm{diag}(\boldsymbol{J}) \odot \boldsymbol{\omega}_n^2, \quad \boldsymbol{k}_\omega = 2\zeta\,\mathrm{diag}(\boldsymbol{J}) \odot \boldsymbol{\omega}_n, \tag{9}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{10}$$

where $\boldsymbol{\omega}_n = (4, 4, 3.5)^{\mathsf{T}}$ rad/s, $\zeta = 0.95$, and $\mathrm{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. Here $\boldsymbol{u}_0$ is a requested body torque, not the final rotor input. The law is a quaternion feedback baseline [9]; its gains give the intended local diagonal second-order approximation. It omits desired-rate feedforward and gyroscopic cancellation. Consequently, tracking a moving reference is evaluated empirically, not covered by a regulation proof.

For perspective, if the reference is constant, $\boldsymbol{k}_q = k_q \mathbf{1}$, no saturation or residual is present, and the trajectory remains on one smooth branch, then $V = \frac{1}{2}\boldsymbol{\omega}^{\mathsf{T}} \boldsymbol{J} \boldsymbol{\omega} + 2k_q(1 - |q_{e,0}|)$ satisfies $\dot{V} = -\boldsymbol{\omega}^{\mathsf{T}}\,\mathrm{diag}(\boldsymbol{k}_\omega)\boldsymbol{\omega}$. This scalar-gain energy identity cannot simply be asserted for the anisotropic, saturated, time-varying experiment in (10).

C. Requested and Realized Wrench

For parallel rotor axes with positions $\boldsymbol{p}_i$ and signed reaction coefficients κ_i, the modeled allocation is

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{11}$$

Assuming nonsingular B, the runner computes

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}}, \\ \boldsymbol{f}_{\mathrm{act}} &= \mathrm{clip}_{[\mathbf{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^{\mathsf{T}}]^{\mathsf{T}} &= B \boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{12}$$

The force units and reaction coefficients come from the model's actuator gears and site geometry. Feasible rotor inputs are enforced by the last clip, but exact realization of the requested wrench requires $\boldsymbol{f}_{\mathrm{req}} \in [\mathbf{0}, \boldsymbol{f}_{\max}]$. A componentwise torque box $\mathcal{U} = \{\boldsymbol{u} : |u_i| \le \ell_i\}$ is not generally contained in the feasible torque slice at a given collective thrust. Control allocation under such constraints is an established problem [10].

Collective thrust is classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0, 2.5Mg]}\left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)} \right). \tag{13}$$

The neural path has no direct collective-thrust output, but it can affect vertical motion through attitude, tilt compensation, and rotor clipping. Rotor feasibility depends jointly on T and $\boldsymbol{u}$.

III. CMG State Dynamics and Online Decoder Adaptation

A. Encoding, States, and Numerical Update

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^{\pm} &= I_b \mathbf{1} + I_g \max(\pm\bar{\boldsymbol{u}}, \mathbf{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^{+})^{\mathsf{T}}, (\boldsymbol{I}^{-})^{\mathsf{T}}]^{\mathsf{T}}. \end{aligned} \tag{14}$$

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{15}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{16}$$

$$c_j = (g_0 + \chi g_1 |I_j|) r_j^2, \qquad \chi \in \{0, 1\}. \tag{17}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance. Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{18}$$

$$r_{j,k}^{+} = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{19}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1 |I_{j,k}|)(r_{j,k}^{+})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k} v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v_{j,k}^{+} = e^{-\lambda_{j,k} h} v_{j,k} + (1 - e^{-\lambda_{j,k} h}) d_{j,k}/\lambda_{j,k}. \tag{20}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v_{j,k}^{+}, r_{j,k}^{+})$.

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G \boldsymbol{r}_k^{+}), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k \boldsymbol{a}_k). \end{aligned} \tag{21}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous

sign agreement after a command reversal. The teaching target and local update are

$$\boldsymbol{y}_k^* = \mathrm{clip}_{[-1,1]}(-k_e \boldsymbol{e}_{q,k} - k_\omega^* \boldsymbol{\omega}_k), \tag{22}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}_k^* - \boldsymbol{s}_k, \tag{23}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta \boldsymbol{\varepsilon}_k \boldsymbol{a}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{24}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m \times 2m}$, and η and λ_W are per-step coefficients, not continuous-time rates. Holding them fixed while changing h changes the learning dynamics. PES learns a prescribed residual target; it does not differentiate through the plant or directly minimize the closed-loop tracking cost. In particular, with a clipped readout this rule is not the exact gradient of the clipped instantaneous squared loss. It is a PES-style error-directed update with projection and leak.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta \boldsymbol{\ell} \odot \boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \qquad 0 \le \beta \le 1. \tag{25}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table I, $r_{j,0} \in [0,1]$, and $0 < h \le \tau_r$, the gate remains in $[0,1]$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. The unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0,1]$. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \ge 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}. Equation (20) is a convex combination of $v_{j,k}$ and d_j/λ_j. Induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \le \beta \ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \le |\beta \ell_i s_{k,i}| \le \beta \ell_i$. The recovery statements follow by substitution. □

The same allocator and state then give the same actuator output when $\beta = 0$. The proposition does not establish state safety, rotor headroom, stability, or a bound on tracking improvement. A bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (24) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta \|\boldsymbol{a}\|_2^2) \boldsymbol{\varepsilon}_k. \tag{26}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta \|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta \boldsymbol{\varepsilon}_k \|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (26). The scalar multiplier has modulus less than one under the stated condition. □

This is the constant-feature PES result [11], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \le 2m$, $0 < \eta < 1/m$ is sufficient for that restricted model with nonzero activity; changing targets, loss of excitation, clipping, and leakage invalidate a direct application to the experiment. No unique identification of all decoder weights follows from one fixed regressor.

D. Annealed Activity-Trace Variant

The archived option called `eprop_anneal` replaces instantaneous presynaptic activity with an exponentially smoothed trace. We call it *annealed activity-trace learning* (AAT), not full e-prop: it does not compute the recurrent state-sensitivity eligibility derivatives of [12]. For $\xi_k = \mathrm{clip}(t_k/T, 0, 1)$,

$$\rho_k = 0.98 - 0.48\xi_k, \qquad \boldsymbol{z}_{-1} = 0, \tag{27}$$

$$\boldsymbol{z}_k = \rho_k \boldsymbol{z}_{k-1} + (1 - \rho_k)\boldsymbol{a}_k, \tag{28}$$

$$\eta_k^{\mathrm{AAT}} = 2\eta_{\mathrm{PES}}(1 - 0.8\xi_k), \tag{29}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta_k^{\mathrm{AAT}} \boldsymbol{\varepsilon}_k \boldsymbol{z}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{30}$$

The factor two is present in the archived constructor. Thus this comparison changes both temporal smoothing and learning-rate schedule. At $h = 0.5$ ms, the instantaneous frozen-ρ equivalent time constant $-h/\log\rho$ falls from about 24.75 ms to 0.72 ms; it does not become exactly instantaneous. Meanwhile the learning rate falls from $2\eta_{\mathrm{PES}}$ to $0.4\eta_{\mathrm{PES}}$.

Proposition 4 (Trace bound and frozen-regressor interpretation)**.** *With $\boldsymbol{z}_{-1} = 0$, $\boldsymbol{a}_k \in [0,1]^{2m}$, and $0 \le \rho_k \le 1$, the trace stays in $[0,1]^{2m}$. Projection keeps AAT decoder weights in $\mathcal{W}$. For fixed target and fixed $\boldsymbol{a}, \boldsymbol{z}$, with no leak, readout clipping, or projection, its error obeys $\boldsymbol{\varepsilon}_{k+1} = (1 - \eta_k^{\mathrm{AAT}} \boldsymbol{z}^{\mathsf{T}} \boldsymbol{a})\boldsymbol{\varepsilon}_k$.*

Proof. Trace boundedness follows from convexity. Decoder boundedness follows from projection. The last identity follows by multiplying the unprojected weight increment by $\boldsymbol{a}$ and subtracting it from the target error. □

Hence alignment between trace and present activity affects learning. This frozen-regressor identity does not establish convergence when the trace, reference, and plant evolve together.

E. Nengo Realization

In Nengo 4.1.0, CMG is a custom non-spiking `NeuronType` with probeable membrane and gate states. Constructed currents are injected into `ensemble.neurons`; conventional NEF gain/intercept fitting is bypassed. The result is a direct-current neural basis with a Nengo PES decoder, not a claim that the full classical controller has been compiled into an NEF population representation.

TABLE I
CANONICAL 3-CHANNEL CMG–PES PARAMETERS. NEURAL STATES/CURRENTS ARE NORMALIZED MODEL VARIABLES; LEARNING COEFFICIENTS ARE PER STEP.

Quantity	Value
h, duration	0.0005 s, 45 s
τ, τ_r	0.05 s, 0.005 s
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$
$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
I_b, I_g, G	$0.75, 1.25, 10^5$
$\beta, \eta, \lambda_W, W_{\max}$	$0.5, 8 \times 10^{-4}, 10^{-6}, 6$
k_e, k_ω^*	$8, 0$

With unfiltered activity and $n = 2m$, Nengo's update scaling is matched using $\eta_N = \eta n/h$ and Nengo error $\boldsymbol{s}_k - \boldsymbol{y}_k^*$. The implementation subtracts $\lambda_W W_k$ using the pre-step weights and projects after the simulator step. The measured worst relative mean-error difference across the ten missions is 2.013×10^{-6}; an isolated command test gives approximately 1.05×10^{-7}. These are numerical comparisons over tested trajectories, not bitwise identity or independent validation of the control hypothesis.

IV. EXPERIMENTAL PROTOCOL AND CROSS-MISSION RESULTS

A. Models, Parameters, and Measurements

MuJoCo supplies rigid-body and actuator simulation [13]. The X2 model uses four rotor actuators. The CF2 case is an explicit four-rotor design with $1.5\times$ nominal upper thrust authority; it is neither stock-Crazyflie hardware validation nor a validated motor upgrade. For both rotor models, $\boldsymbol{\ell} = \max(\mathrm{diag}(J) \odot (12, 12, 10)^{\mathsf{T}}, 10^{-6}\mathbf{1})$ in N m, with the maxima taken componentwise.

Each baseline/CMG pair uses the same initial pose/rate, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic, with no injected disturbances or sensor noise. One run per cell is reported; repeated numerical integration of the same deterministic case is not statistical replication. Learning remains active throughout evaluation. We report the sample mean $\bar{\theta}_q$ and relative reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$. Summary metrics use all 90,000 integration samples; most saved plotting traces retain every tenth sample (5-ms spacing), while the X2 Hard45 trace is full-rate. Thus recomputing a mean from the decimated plot data need not exactly reproduce the full-rate metric.

The five reference triples are generated as follows (radians): Roll is $(0.25 \sin 0.7t, 0, 0)$; Full3 is $0.5 \sin(t + (0, \pi/2, \pi)^{\mathsf{T}})$, componentwise. Hard45 is

$$\begin{bmatrix} 0.55\sin(0.75t) + 0.20\sin(2.10t + 0.40) \\ 0.45\sin(0.95t + \pi/3) + 0.18\sin(1.75t) \\ 0.65\sin(0.55t + \pi) + 0.25\sin(1.35t + 0.20) \end{bmatrix}. \tag{31}$$

TABLE II
CANONICAL NENGO CMG–PES VERSUS PD: MEAN QUATERNION-ANGLE ERROR [RAD], 45 S, $n = 1$ PER CELL. S IS THE LOGGED PERCENTAGE OF SAMPLES WITH AT LEAST ONE ROTOR COMMAND AT A BOUND, PD/CMG. CF2 DENOTES THE $1.5\times$ ROTOR-AUTHORITY DESIGN, NOT STOCK HARDWARE.

Model	Mission	PD	CMG–PES	Reduction	S [%]
X2	Roll	.050565	.041785	17.36%	0/0
	Full3	.303275	.131748	56.56%	0/0
	Hard45	.294082	.138007	53.07%	0/0
	Hardstep	.454528	.359355	20.94%	.016/.239
	Yaw360	.075863	.029902	60.58%	0/0
CF2	Roll	.051337	.042439	17.33%	0/0
	Full3	.288383	.126476	56.14%	0/0
	Hard45	.285449	.132319	53.65%	0/0
	Hardstep	.426182	.347886	18.37%	.137/.264
	Yaw360	.075812	.029870	60.60%	0/0

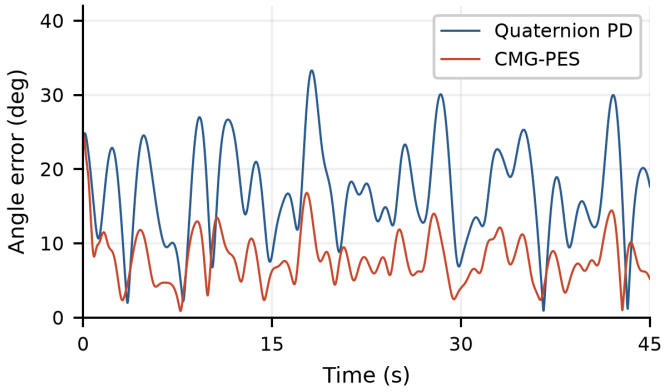


Fig. 1. X2 Hard45 quaternion-angle error over the complete matched 45-s run. The Nengo CMG–PES mean reduction is 53.07%; adaptation remains active during evaluation. This panel shows angle error, not three Euler tracking trajectories.

Hardstep holds zero for one second, then cycles every two seconds through the rows of

$$\begin{bmatrix} .75 & -.45 & .55 \\ -.65 & .55 & -.70 \\ .50 & .50 & .90 \\ -.80 & -.35 & .20 \end{bmatrix}. \tag{32}$$

Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. The nominal Hover equilibrium is also available, but its essentially zero error does not support a meaningful percentage comparison.

B. Primary Five-Mission Matrix

All ten primary cases improve mean error (Table II). Gains are largest on Yaw360 and Full3 and smallest on Roll; the abrupt Hardstep case is intermediate. The reported bound-occupancy statistic detects post-clipping commands within the runner's tolerance of a limit. It is not a measurement of requested-wrench infeasibility magnitude. Hardstep has small but nonzero occupancy on both models, so a blanket zero-saturation claim would be incorrect. These cases demonstrate repeatable nominal performance, not robustness to model mismatch or a universally optimal PD comparison.

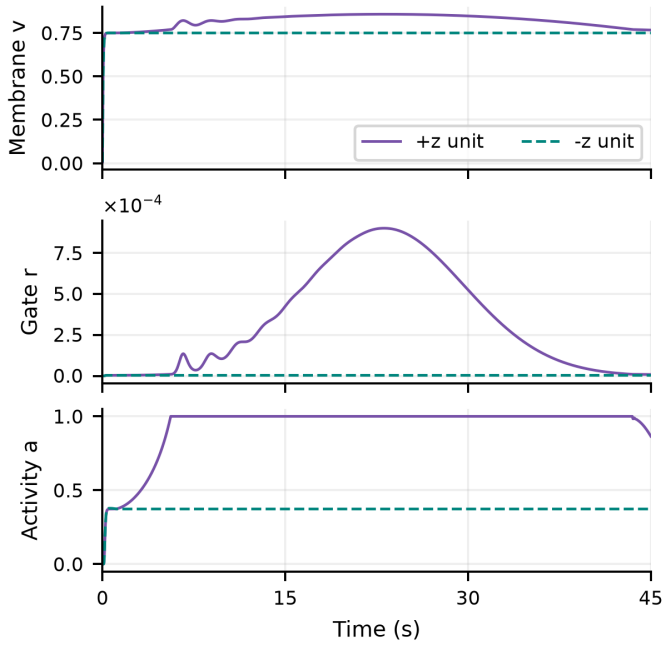


Fig. 2. X2 Yaw360: recorded membrane, gate, and decoder activity of the two yaw-current units (the other four units are omitted). A gate much smaller than one can saturate the amplified activity. These are continuous states, not spikes or physical voltage measurements.

C. What Is Learning?

Fixed-decoder residuals already improve some missions: in the audited direct X2 runs their mean reductions range from about 0.12% to 10.27%; the corresponding CF2-design range is 0.06% to 6.80%. Thus it is inaccurate to say that fixing the decoder removes *all* improvement. PES produces the much larger 17.3–60.6% primary reductions. Separately, toggling χ with the decoder mode held fixed is numerically indistinguishable under the tested settings. This identifies decoder adaptation as the principal tested lever, not a universal claim that conductance never matters.

Because $a_j = \mathrm{clip}(10^5 r_j, 0, 1)$, decoder activity saturates already at $r_j = 10^{-5}$. Tiny but nonzero gates can therefore yield order-one activity while $c_j \propto r_j^2$ remains small. This is a plausible mechanistic explanation for the observed conductance insensitivity, not a claim that every gate remains $O(10^{-6})$ at every instant. A calibrated gate-gain sweep and gate/membrane readout ablation are needed to determine whether CMG offers benefits beyond a simpler dynamical basis.

Figure 2 distinguishes the membrane, raw gate, and clipped activity using the same recorded trajectory rather than equating them.

D. Neuron-Family Comparison

The Nengo benchmark contains two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG, LIF, ALIF, Izhikevich [14], and LIFRate. All use the same residual/allocator path, with model-specific native activity and filtering. Some Izhikevich–PES cells outperform CMG–PES on mean tracking error. The audit reports an approximate average normalized residual RMS of 0.37 for Izhikevich–PES versus 0.12 for CMG–PES, alongside more decoder movement. Equal output caps do not imply equal realized correction effort. Very sparse LIF/ALIF activity in some tested cells is an operating-point limitation, not evidence that those neuron families cannot learn. Accordingly, the supported conclusion is a tracking–effort tradeoff, not universal CMG superiority or a proven causal effect of spiking.

TABLE III
TRACEABLE PRECURSOR SUMMARIES: MEAN QUATERNION-ANGLE ERROR [RAD], 45 S, $h = 0.5$ MS, ONE RUN PER CONFIGURATION. AAT CORRESPONDS TO ARCHIVED `EPROP_ANNEAL`. THESE RESULTS MUST NOT BE POOLED WITH TABLE II.

Model	Mission	PD	PES	AAT
X2	Hard45	.294079	.122051	.127526
CF2	Hard45	.285448	.035549	.035445
CF2	Hardstep	.661866	1.757814	2.079473

V. Annealed Learning: Verified Precursor Evidence

The verified AAT results use the precursor *blend* law $(1-\beta)\boldsymbol{u}_0 + \beta\boldsymbol{\ell}\odot\boldsymbol{s}$ at $\beta = 0.5$, with ideal external-wrench injection and a classical horizontal stabilizer. Its CF2 torque limits also differ from the rotor-design campaign. Therefore these results are a within-precursor learning-configuration comparison, not additional rotor-feasible validation of (25).

On Hard45, AAT is slightly better than PES for CF2 and worse for X2. Both learned configurations severely degrade CF2 Hardstep tracking relative to that precursor baseline. Since quaternion angle is bounded by π, these values indicate a loss of tracking, not mathematical divergence of the angle or decoder weights. The archived Hardstep saturation field is zero and the actuation path is ideal wrench; these files do not establish rotor-allocation saturation as the cause.

The comparison changes eligibility smoothing and learning-rate schedule together, and the canonical experiment additionally changes residual law and actuator interface. It therefore does not isolate which change avoids the precursor failure. At fixed $\beta = 0.5$, the blend retains a known baseline coefficient of 0.5, not an arbitrarily vanishing coefficient. The additive law removes this explicit attenuation but cannot by itself guarantee stability or eliminate adverse step responses. An additive-versus- blend factorial study with matched actuation, teacher, and learning rate is the appropriate next test.

VI. Selected Preliminary Physical Tests

Before the present manuscript synthesis, CMG–PES was tested on the Quanser 3DOF Hover. These are preliminary supporting tests, not free-flight validation of either MuJoCo airframe. The rig uses torque-to-motor-voltage allocation, a 200-Hz Q8 loop, and ten 0.5-ms Nengo substeps per hardware sample. The representative Roll configuration uses $\beta = 0.2$, a

0.2-V per-motor residual cap, 1 V/s residual slew limit, $\eta = 2 \times 10^{-4}$, $k_e = 2$, and $k_\omega^* = 0$.

The strongest replicated result is Roll: over three matched 8-s pairs, the recorded mean error is $5.4405 \pm 0.0648^\circ$ for PD and $5.2960 \pm 0.0428^\circ$ for CMG–PES. All three pairs improve, giving a paired mean reduction of $2.65\% \pm 1.08\%$ (sample SD, not a confidence interval). These archived values use the mean norm of the wrapped yaw/pitch/roll error vector, not $\bar{\theta}_q$; quaternion errors are logged separately. They are not numerically pooled with the MuJoCo table.

Selected later single pairs give mean-error reductions of 0.55% for Hard45 and 0.95% for a scaled, transition-smoothed Hardstep. Their peak errors worsen, so these are not improvements in every metric. These selected favorable examples do not characterize the full physical campaign: other profiles showed regressions, and no complete full-scale yaw360 pair was validated. The complete record is retained in the companion audit. Hardware supports execution feasibility and a small repeated Roll benefit; it does not identify why the simulation-to-hardware improvement differs.

VII. Discussion and Conclusion

The main result is a transparent architecture-level combination: paired continuous CMG dynamics, local online decoder adaptation, and a bounded additive torque correction around a fixed quaternion-PD law. The five-mission matrix shows 17.3–60.6% lower nominal mean attitude error on two modeled platforms, with explicit actuator-bound accounting. The Nengo realization reproduces the direct computation, and the ablations separate conductance modulation from the stronger decoder-learning effect. AAT provides a well-defined alternative memory/schedule mechanism, but its verified results remain precursor evidence rather than a canonical cross-mission advantage.

Command boundedness, bounded internal states, and projected weights are useful implementation guarantees, not closed-loop stability or safety proofs. An ISS or Lyapunov analysis would additionally require a suitable baseline robustness property and a bound on the *realized* perturbation after allocation. Small voltage caps alone do not quantify achievable tracking improvement. The present nominal gains may partly compensate phase/amplitude errors of the chosen feedforward-free PD, so comparison with stronger classical and simple adaptive residual baselines is important.

Priority follow-up experiments are an additive AAT/PES comparison with matched learning rates, equal-effort neuron/readout ablations, stochastic disturbance trials, and frozen-weight tests on held-out references. Repeated hardware perturbation tests and neuromorphic-device energy/latency measurements would extend, rather than retroactively strengthen, the present evidence. The preliminary Hover Roll result is encouraging, but the paper's central claim remains the neuromorphic residual mechanism and its controlled, cross-mission simulation evaluation.

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